ECEN 454 - Lab Report

Lab Number: 6

Lab Title: Design & Characterization of a Flip-flop

Section Number: 341

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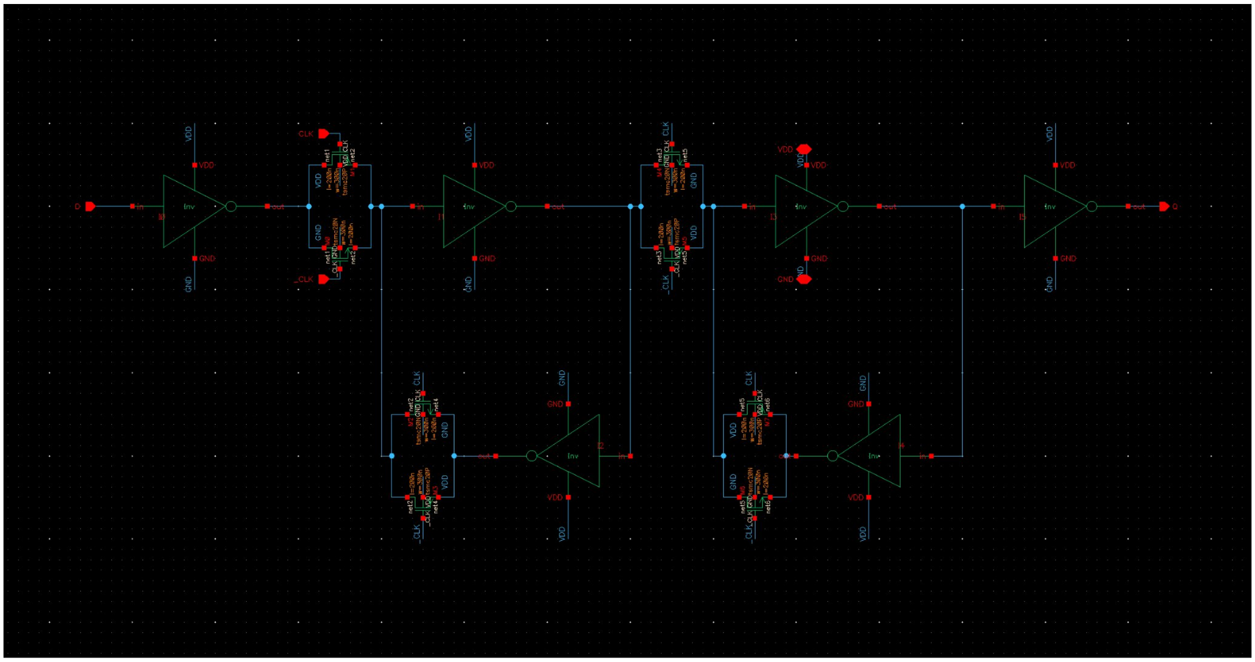
Date: 10/31/2024

(Happy Halloween!)

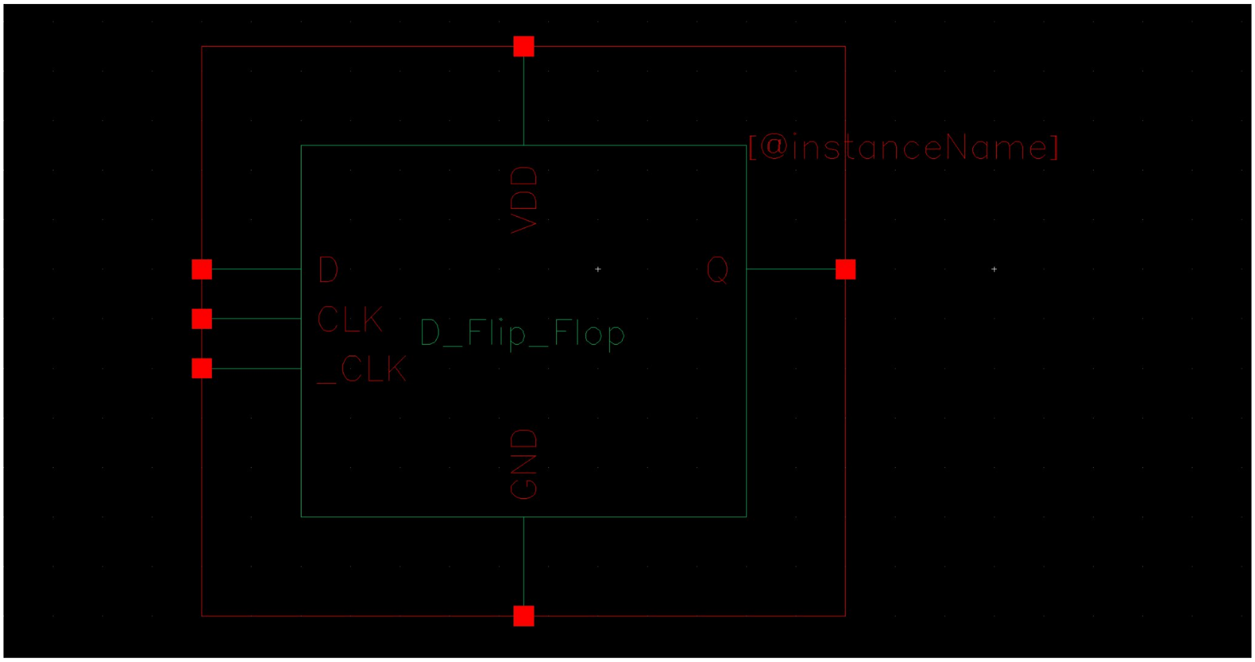
TA: Nishank Bansal

1. <u>FlipFlop Design</u>

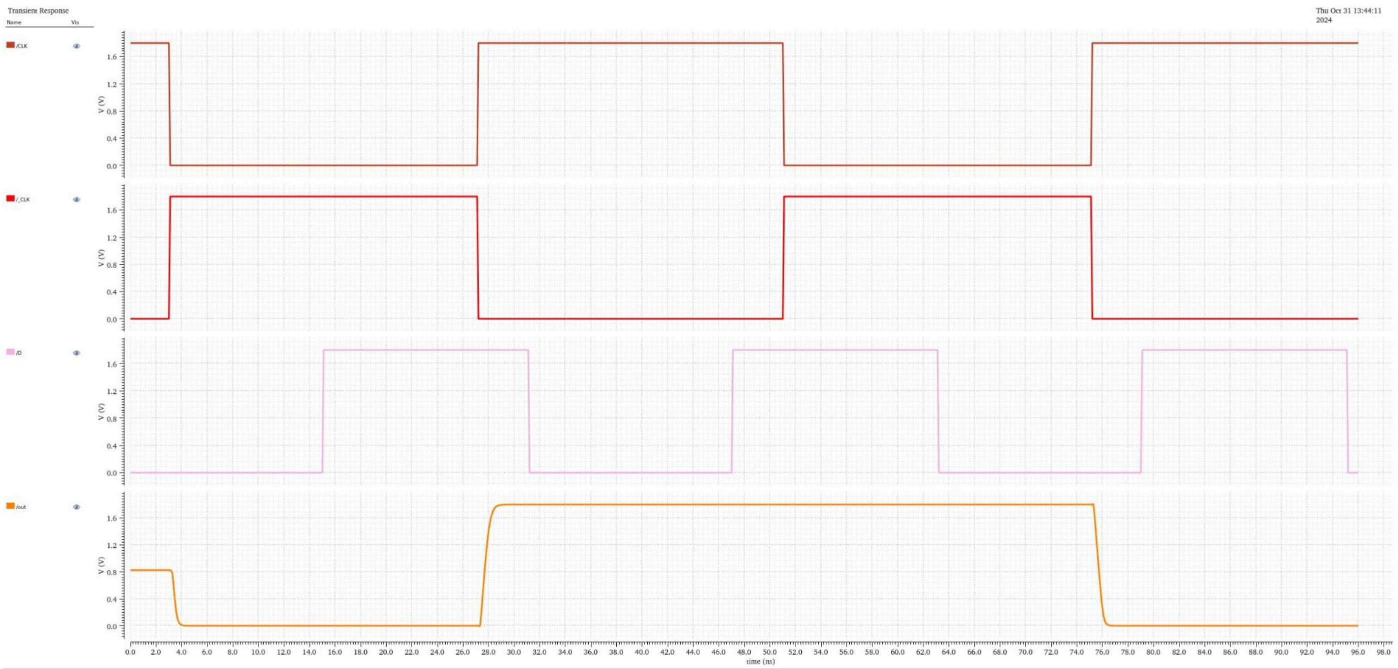
a. Schematic



b. Symbol

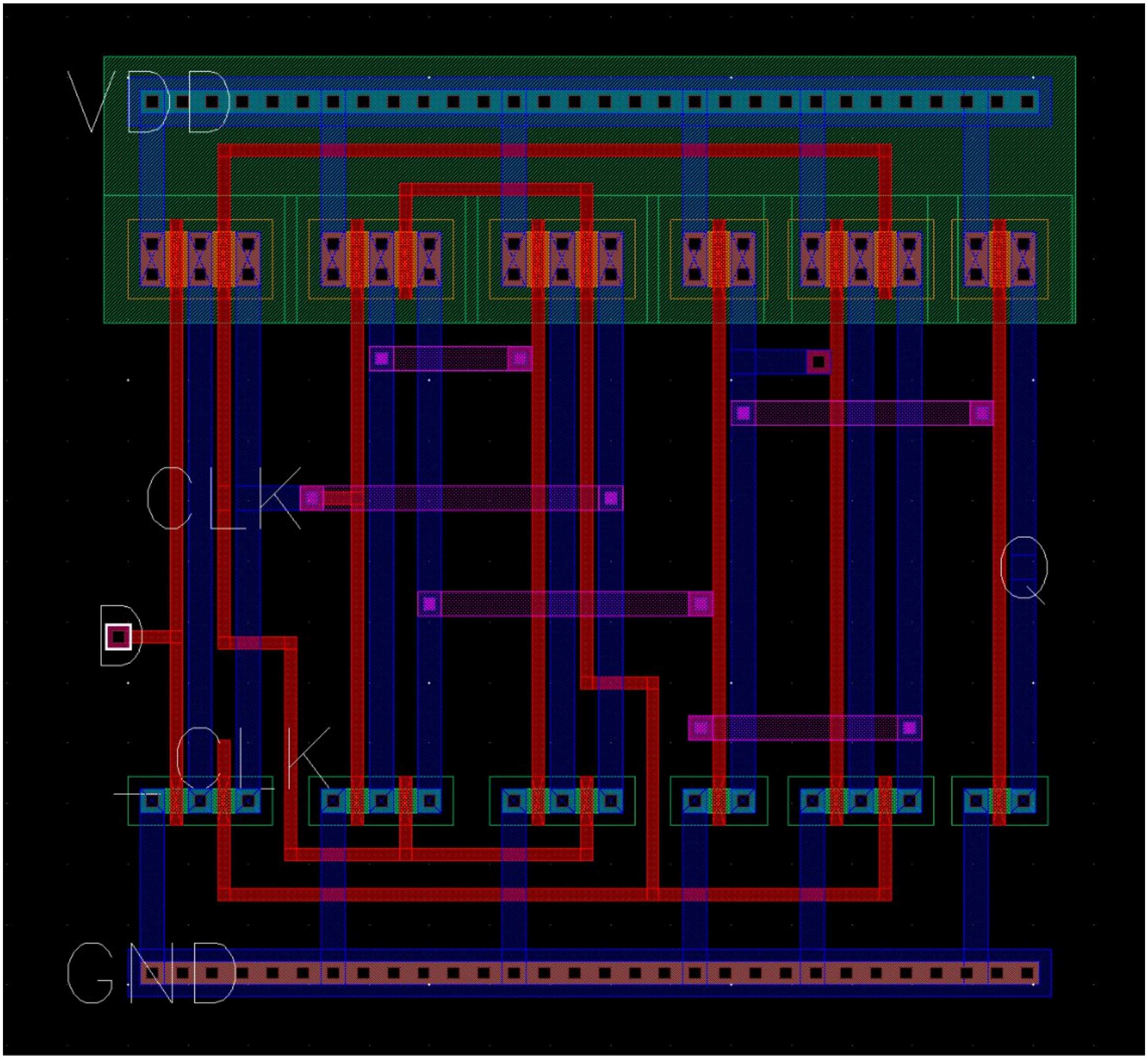


c. Pre-layout Sim Results

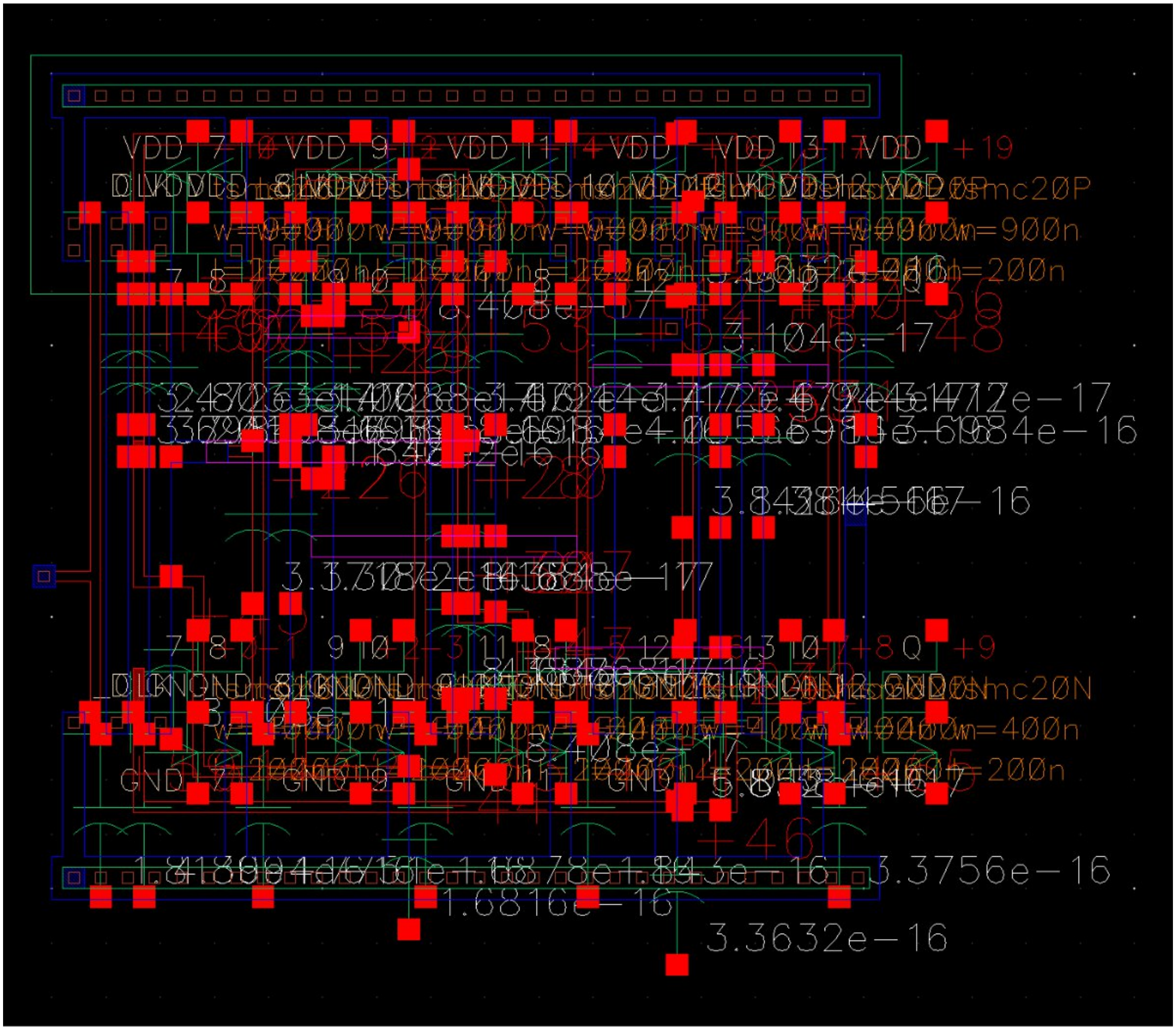


2. FlipFlop Layout

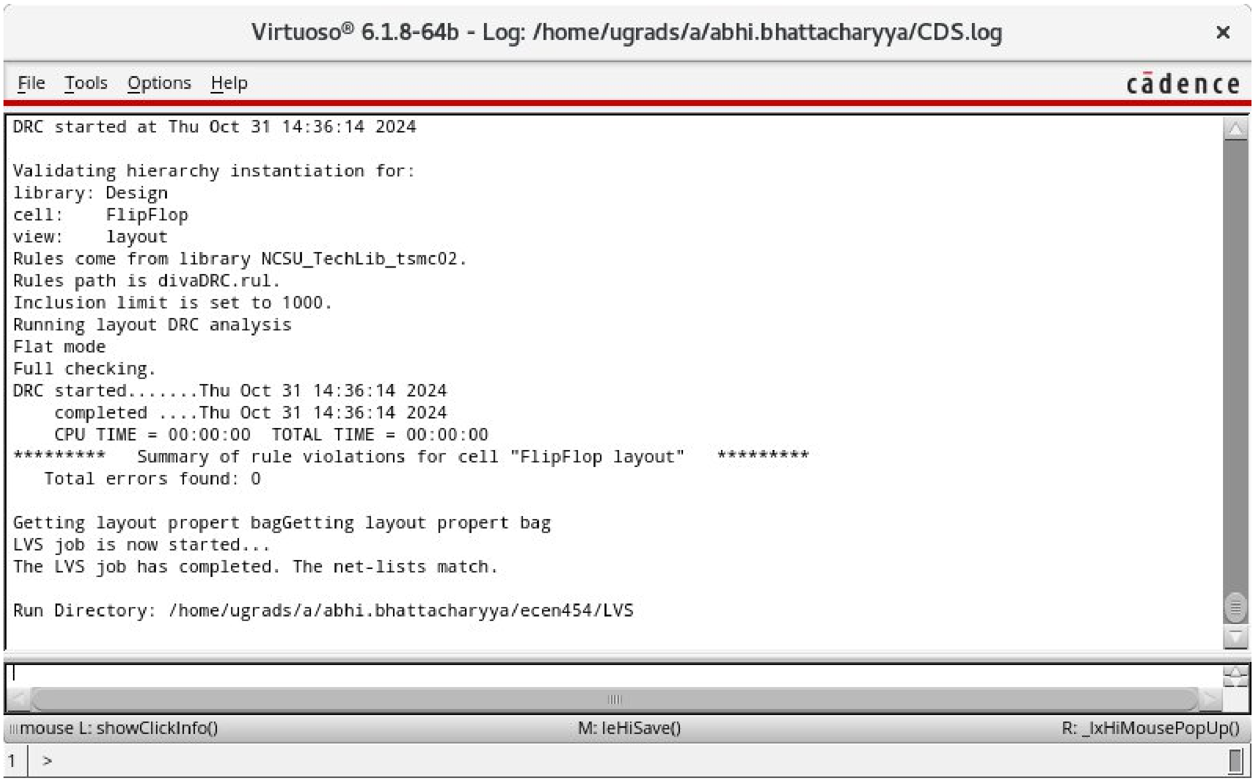
a. Layout



b. Extracted

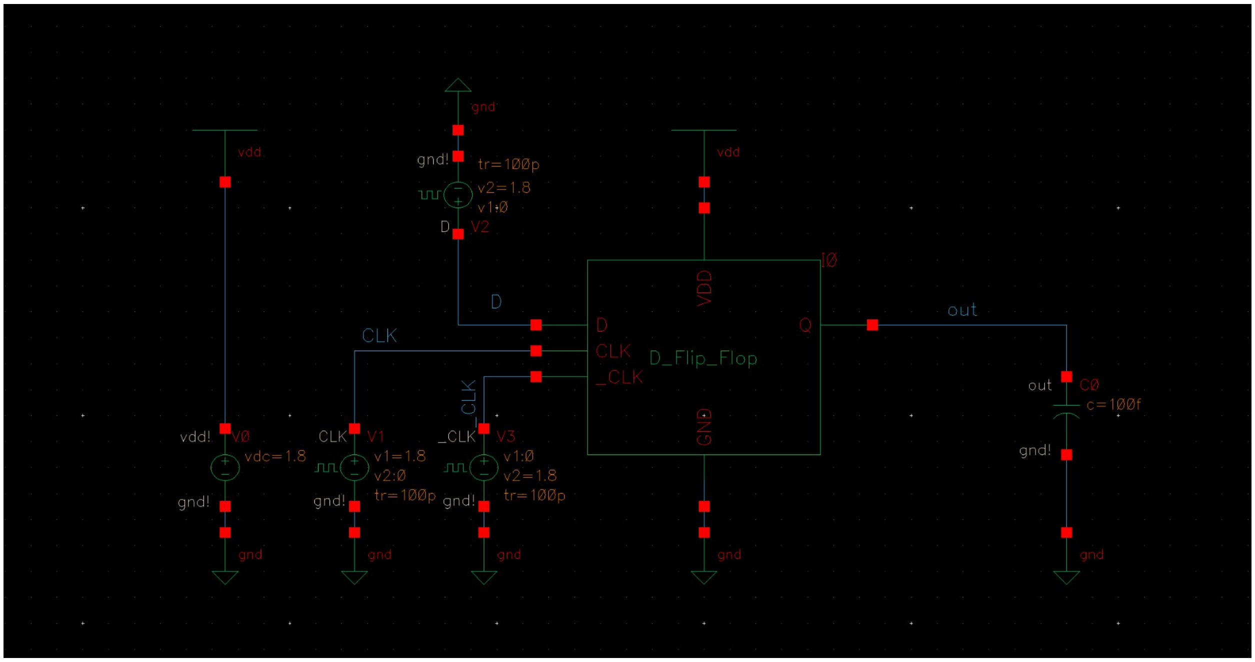


3. DRC/LVS Results

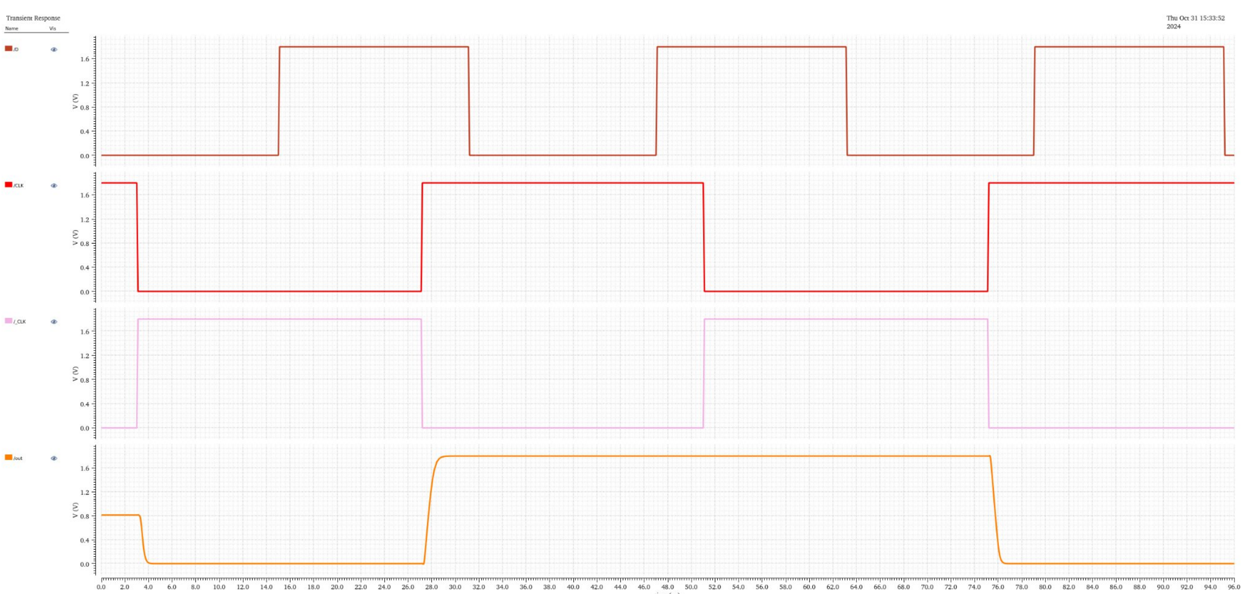


4. Simulation

a. Circuit Schematic

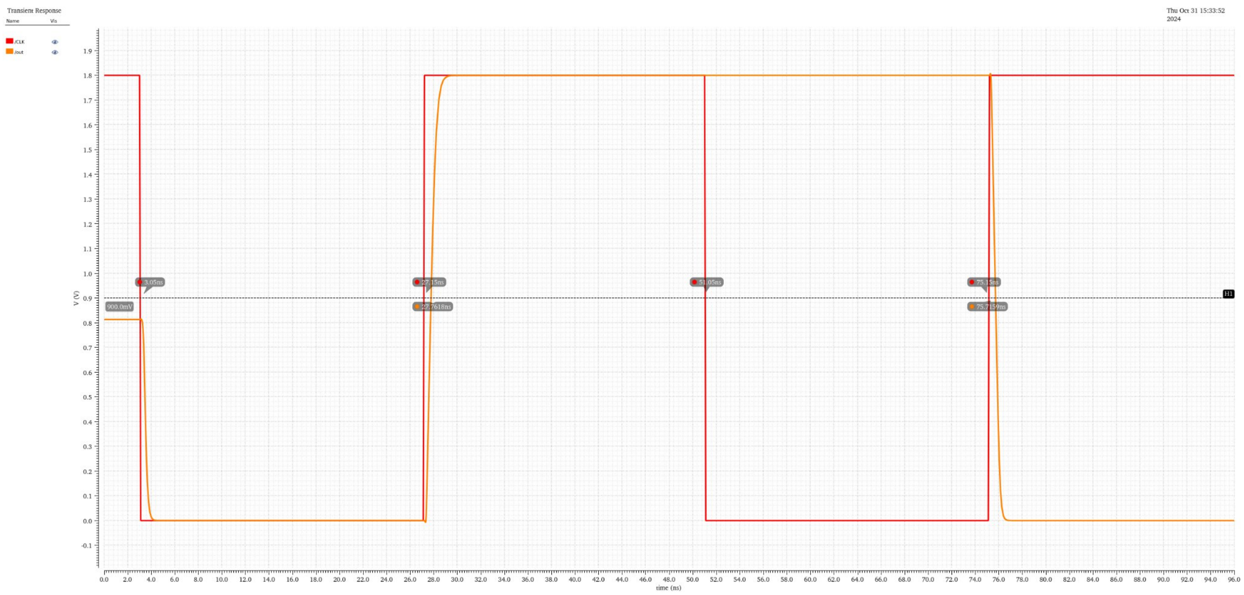


b. Output waveform for every case

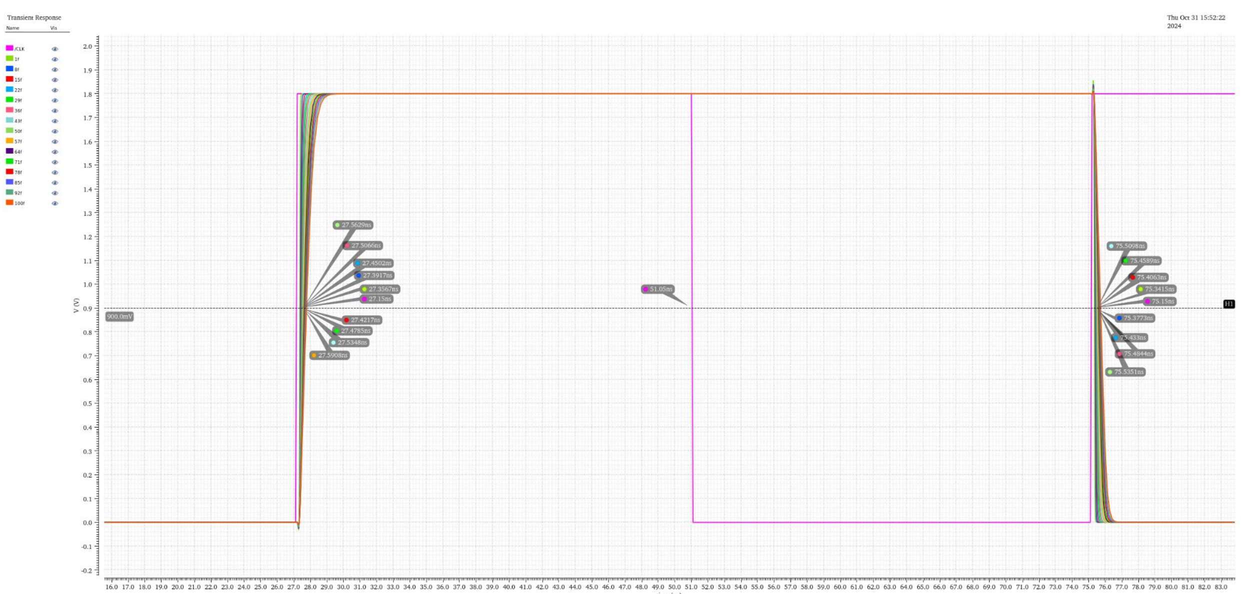


5. Delays

a. Waveform for rise/fall delay time at 100fF



b. Combined waveforms for all rise/fall graphs

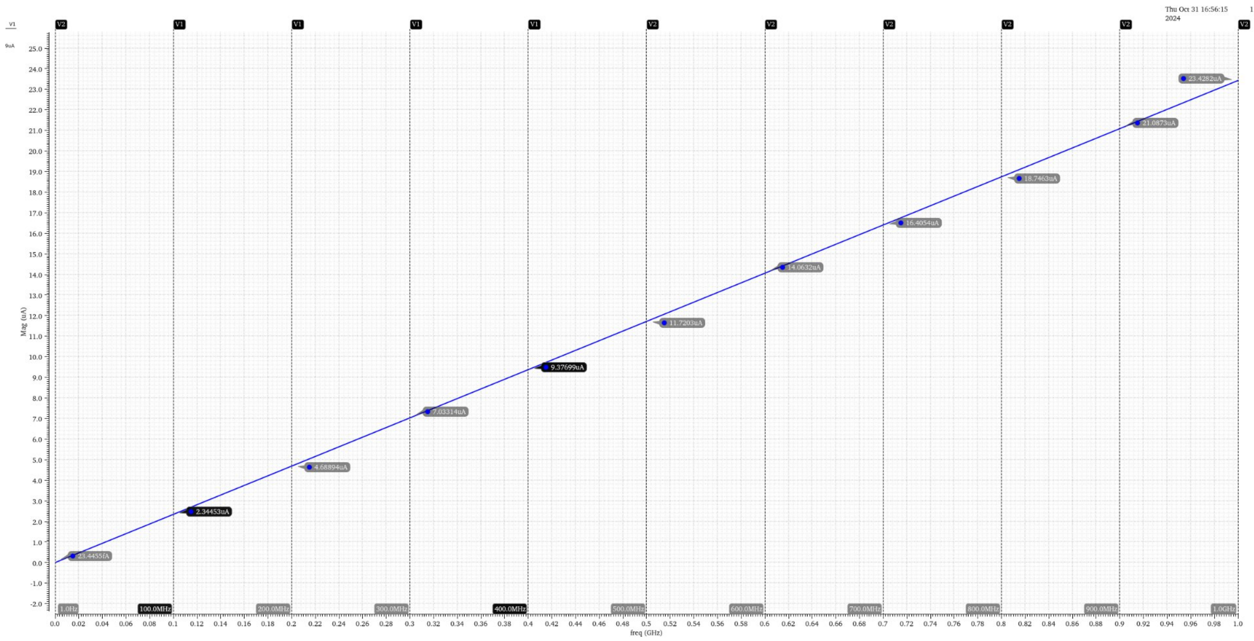


c. Rise/fall delay table

Capacitance (fF)	Input Rise Time	Output Rise Time	Input Fall Time	Output Fall Time	Rising Delay	Falling Delay	% Error
1	27.15	27.3567	75.15	75.3415	0.2067	0.1915	7.94%
8	27.15	27.3917	75.15	75.3773	0.2417	0.2273	6.34%
15	27.15	27.4217	75.15	75.40063	0.2717	0.25063	8.41%
22	27.15	27.4502	75.15	75.433	0.3002	0.283	6.08%
29	27.15	27.4785	75.15	75.4589	0.3285	0.3089	6.35%
36	27.15	27.5066	75.15	75.4844	0.3566	0.3344	6.64%
43	27.15	27.5348	75.15	75.5098	0.3848	0.3598	6.95%
50	27.15	27.5629	75.15	75.5351	0.4129	0.3851	7.22%
57	27.15	27.5908	75.15	75.5605	0.4408	0.4105	7.38%
64	27.15	27.6184	75.15	75.5857	0.4684	0.4357	7.51%
71	27.15	27.6467	75.15	75.611	0.4967	0.461	7.74%
78	27.15	27.6742	75.15	75.6364	0.5242	0.4864	7.77%
85	27.15	27.7023	75.15	75.6615	0.5523	0.5115	7.98%
92	27.15	27.7304	75.15	75.6867	0.5804	0.5367	8.14%
100	27.15	27.7618	75.15	75.7159	0.6118	0.5659	8.11%
w_n = 0.4u					w_p = 1u		

6. Input capacitance from AC Simulation

a. Graph of Frequency analysis w/ 10 points



b. Table of points w/ sink capacitance

Frequency (f)	Current (I)	Sink Capacitance
1	2.34E-14	3.73E-15
1.00E+08	2.34E-06	3.73E-15
2.00E+08	4.69E-06	3.73E-15
3.00E+08	7.03E-06	3.73E-15
4.00E+08	9.38E-06	3.73E-15
5.00E+08	1.17E-05	3.73E-15
6.00E+08	1.41E-05	3.73E-15
7.00E+08	1.64E-05	3.73E-15
8.00E+08	1.87E-05	3.73E-15
9.00E+08	2.11E-05	3.73E-15
1.00E+09	2.34E-05	3.73E-15
	Avg	3.73E-15

c. **Average sink capacitance = 3.73E-15 = 3.73 Ff**

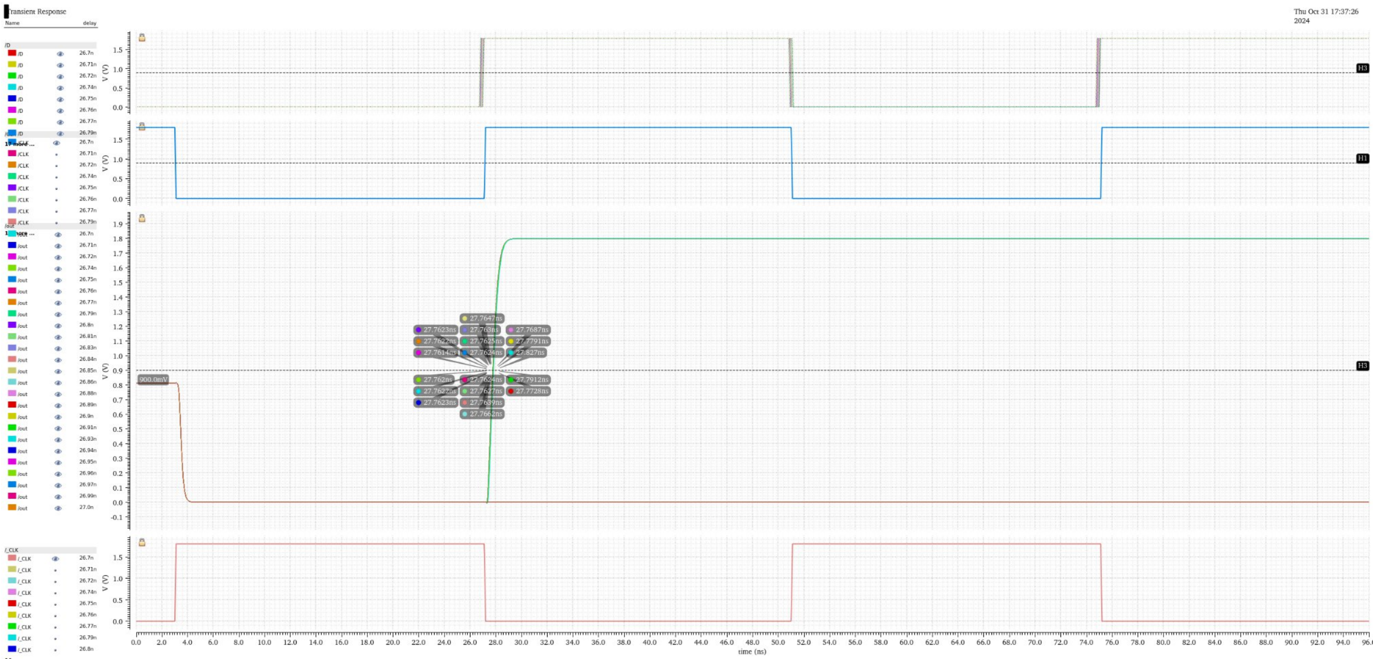
7. Setup times

a. Methodology Explained

I found the setup times using the instructions for parametric simulations given in the lab report. For rising time, I set up the input signal D such that the signal starts low, and then goes high just before the rising edge of the clock. I first tested a data range between 26.5 ns - 26.75 ns, which is between a 0.25ns - 0.5ns setup time. This did not reveal anomalous data for the output Q, so I re-tested with the range from 26.7 ns to 27.0 ns, which showed a setup time of 0.32ns for rising. THe times in “D Rise Time” are not the value of the delay variable; I instead used a marker at 0.9V for the clock, Q, and D rise time to ensure these are all consistent with each other.

The first behavior I saw for both rising and falling setup was “a dramatic increase in Clock-to-Q delay”. I visualized this by adding a column for Δ Delay, the increase in delay between each iteration. I defined a “dramatic increase” as the delay going up by more than 0.0005ns between two iterations. For the rising edge, this occurred at rising time = 26.84ns, where delay changed from 0.0003 on the last iteration to 0.0009 in that iteration. The delay then dramatically increased by 0.0015 at rising time = 26.86ns. I considered the setup time at 0.32ns there is only a 0.03ns difference and it is better to have a safety margin when finding the setup time specification for an IC to ensure it works as specified in the field and non-ideal conditions. I did this very similarly for the falling edge and found a setup time of 0.29ns. This made the overall setup time (the worst between the two) **0.32ns.**

b. Rising setup waveforms

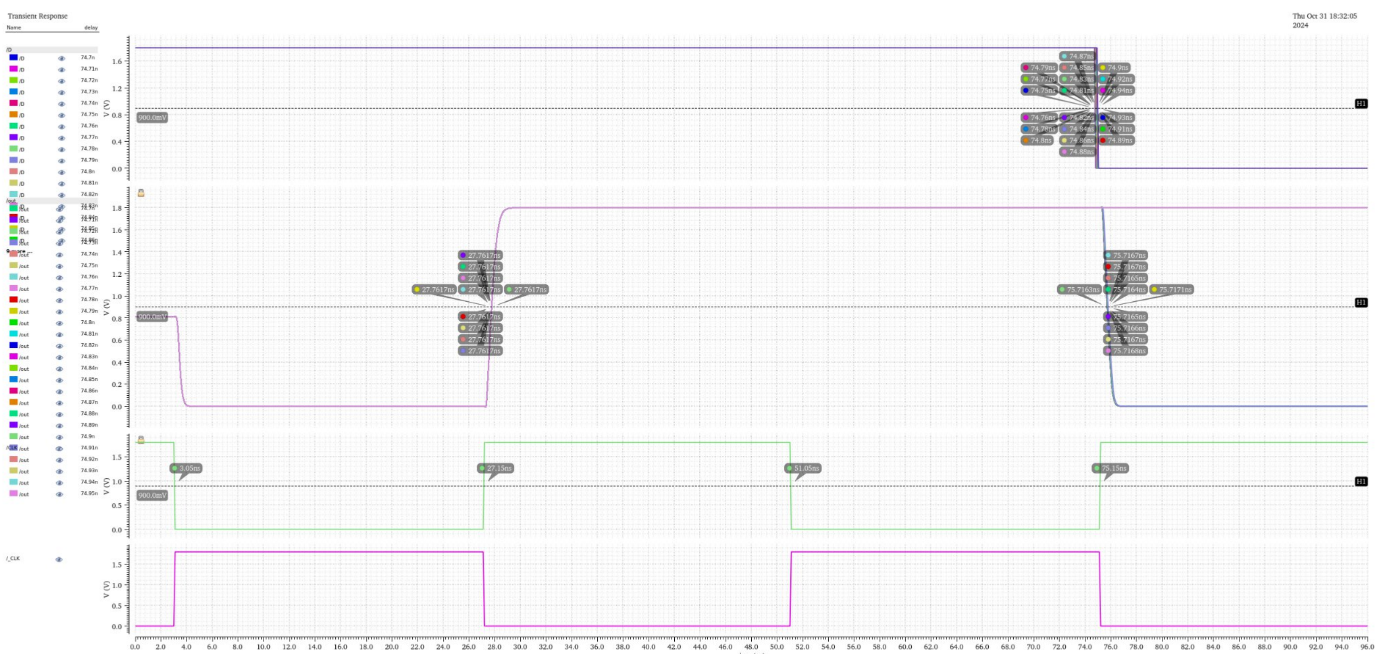


c. Rising setup table

Rising Setup Time (all times in ns)					
Clock Rise TIme	Q Rise TIme	D Rise Time	Clock-to-Q Delay	Δ Delay	Setup Time
27.15	27.7622	26.7	0.6122	---	0.45
27.15	27.7623	26.71	0.6123	0.0001	0.44
27.15	27.7614	26.72	0.6114	-0.0009	0.43
27.15	27.762	26.74	0.612	0.0006	0.41
27.15	27.7624	26.75	0.6124	0.0004	0.4
27.15	27.7624	26.76	0.6124	0	0.39
27.15	27.7622	26.77	0.6122	-0.0002	0.38
27.15	27.7625	26.79	0.6125	0.0003	0.36
27.15	27.7623	26.8	0.6123	-0.0002	0.35
27.15	27.7627	26.81	0.6127	0.0004	0.34
27.15	27.763	26.83	0.613	0.0003	0.32
27.15	27.7639	26.84	0.6139	0.0009	0.31

27.15	27.7647	26.85	0.6147	0.0008	0.3
27.15	27.7662	26.86	0.6162	0.0015	0.29
27.15	27.7687	26.88	0.6187	0.0025	0.27
27.15	27.7728	26.89	0.6228	0.0041	0.26
27.15	27.7791	26.9	0.6291	0.0063	0.25
27.15	27.7912	26.91	0.6412	0.0121	0.24
27.15	27.827	26.93	0.677	0.0358	0.22
27.15	doesn't rise	26.94			0.21
27.15	doesn't rise	...			
27.15	doesn't rise	27			
			Rising Setup Time:		**0.32 ns**

d. Falling setup waveforms



e. Falling edge table

Falling Setup Time (all times in ns)					
Clock Rise TIme	Q Fall TIme	D Fall Time	Clock-to-Q Delay	Δ Delay	Setup Time
75.15	75.7164	74.75	0.5664	---	0.4

75.15	75.7165	74.76	0.5665	0.0001	0.39
75.15	75.7163	74.77	0.5663	-0.0002	0.38
75.15	75.7166	74.78	0.5666	0.0003	0.37
75.15	75.7165	74.79	0.5665	-0.0001	0.36
75.15	75.7167	74.8	0.5667	0.0002	0.35
75.15	75.7167	74.81	0.5667	0	0.34
75.15	75.7168	74.82	0.5668	0.0001	0.33
75.15	75.7167	74.83	0.5667	-0.0001	0.32
75.15	75.7171	74.84	0.5671	0.0004	0.31
75.15	75.7172	74.85	0.5672	0.0001	0.3
75.15	75.7176	74.86	0.5676	0.0004	0.29
75.15	75.7177	74.87	0.5677	0.0001	0.28
75.15	75.7185	74.88	0.5685	0.0008	0.27
75.15	75.7193	74.89	0.5693	0.0008	0.26
75.15	75.7205	74.9	0.5705	0.0012	0.25
75.15	75.7219	74.91	0.5719	0.0014	0.24
75.15	75.7245	74.92	0.5745	0.0026	0.23
75.15	75.7289	74.93	0.5789	0.0044	
75.15	75.7363	74.94	0.5863	0.0074	
75.15	75.7505	74.95	0.6005	0.0142	
75.15	75.7885	74.96	0.6385	0.038	
75.15	doesn't fall	74.97			
75.15	doesn't fall	...			
75.15	doesn't fall	75			
			Falling Setup Time:		**0.28 ns**

f. Overall Setup Time (longe of the two): 0.32 ns